

LEMS-A3: THE LUNAR ENVIRONMENTAL MONITORING STATION—AN ARTEMIS III DEPLOYED INSTRUMENT. D. DellaGiustina¹, N. Schmerr², M. Benna^{3,4}, N. McCall⁴, A. Marusiak¹, V. J. Bray¹, S. Bailey¹, P. K. Byrne⁵, B. Avenson⁶, D. Kim⁶, and the Artemis III Science Team. ¹University of Arizona (danidg@orex.lpl.arizona.edu), ²University of Maryland, College Park³, University of Maryland, Baltimore County, ⁴NASA Goddard Space Flight Center, ⁵Washington University in St. Louis, ⁶Silicon Audio Inc.

Introduction: A detailed picture of the Moon's interior has evolved from measurements taken during the Apollo era. The Apollo Lunar Surface Experiment Packages (ALSEPs) formed a four-station passive seismic network that was deployed by astronauts on the nearside of the Moon and returned the first-ever seismic dataset for another body in the Solar System [1]. The ALSEP seismometers established that the Moon is a seismically active body, recording over 13,000 moonquakes [2, 3] over the seven years the network was operational. Studies of the Apollo moonquake dataset revolutionized our understanding of the lunar interior, providing estimates for lunar crustal thickness and layering beneath the stations [4–6], a preliminary seismic wave velocity and the bulk structure and composition for the lunar mantle [7, 8], and estimates for the size, state, and composition of the lunar core [9, 10].

Despite these accomplishments, our current understanding of the lunar interior is biased towards the nearside, equatorial, geographic location of the ALSEP network, with few detections of farside seismicity and seismic observations situated primarily above the structure of the anomalous nearside Procellarum KREEP Terrane (PKT). The Lunar Environment Monitoring Station for Artemis III (LEMS-A3) is a selected Deployed Instrument for Artemis III (A3), that when deployed near the lunar South Pole at the margin of the South Pole–Aitken basin (SPA), will provide new insights into a lunar location that remains seismically unexplored.

Mission Design: LEMS-A3 is a compact, autonomous, and self-sustaining seismometer suite designed to carry out continuous (day and night), long-term, monitoring of the lunar seismic environment at the South Polar region (**Fig. 1**). Once deployed on the surface and activated by the A3 crew, it requires no support (power, thermal, commanding, or data) from the Human Landing System (HLS)/Starship or the A3 crew to operate. The LEMS-A3 geophysical station (**Fig. 1**) continues operations for a targeted two-year lifetime goal, long after the A3 crew departs the Moon, providing a long-lived geophysical surface observatory.

LEMS-A3 has three elements: (a) a three-axis Short Period Seismometer (SP); (b) a complementary three-axis Broad Band Seismometer (BB); and (c) an Instrument Suite Platform (ISP) that provides data digitization and storage, power, thermal, and communication resources. Subsurface burial of the of the LEMS-A3 seismometers into the lunar regolith by the A3 crew will provide thermal stability, improved coupling to the ground, and reduced scattered seismic noise, all issues that plague surface-emplaced seismometer packages.

From the perspective of LEMS-A3 operations, *any* of the identified A3 candidate landing sites will be suitable for providing direct-to-Earth communications and sunlight for solar power, although optimal deployment sites may exist within each of the down-selected sites.

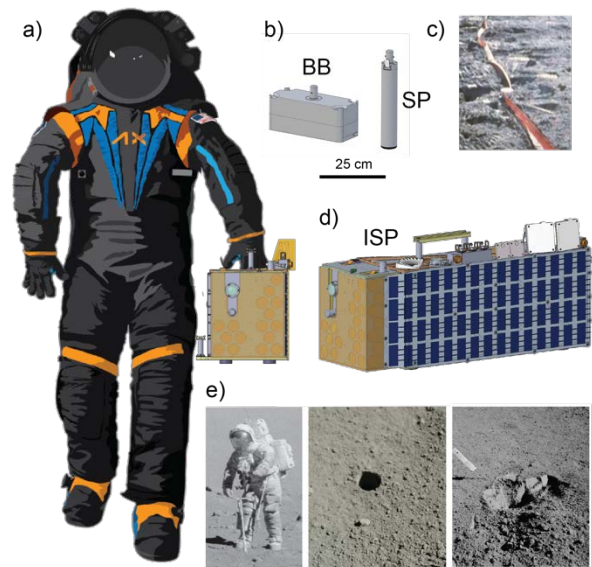


Figure 1. The LEMS-A3 station design and deployment; astronaut and instruments are to scale. a) LEMS-A3 is hand-carried by an A3 astronaut crew member and positioned on the lunar surface using a Sun compass to ensure capture of solar energy and communication directly back to Earth. b) The seismometer suite; instruments are stowed and deployed to the surface via c) a set of ribbon cables, shown here for Apollo, that carries signals to d) the ISP, which houses the avionics and battery, and allows the LEMS-A3 seismometers to operate continuously throughout the lunar day and night. e) A notional A3 crew seismometer deployment strategy, with examples from Apollo operations; SP will be deployed into a core-extracted borehole and BB will be buried in a trench dug for geological sample collection.

Deployment Testing: As LEMS-A3 will be installed by the astronaut crew, we conducted a preliminary series of deployment simulations in a University of Central Florida (UCF) Exolith Lab test chamber. The team studied the following three elements in the simulant chamber under artificial low-angle lighting conditions expected at the South Pole: 1) how to deploy the ISP into a highland simulant that replicates some of the mechanical properties of regolith; 2) the emplacement of the SP seismometer into a borehole using a mockup of the Astronaut Extension Tool (AET); and 3) the

trenching and burial of the BB seismometer (**Fig. 2**). A lunar weight, -shape, and -volume model of the ISP was built for the test, as well as models of the SP, BB, and AET that will be used by the A3 crew. Low-fidelity geological tools were used for the borehole and trench. On the basis of these tests, we established that, before burial, the crew will position and level the BB in the trench and capture its orientation with their camera. For the SP burial in the borehole, the crew member will need to capture the orientation of the AET before detaching it from the seismometer. At the Exolith Lab we also sought to demonstrate how photogrammetry of the deployment site can be used to reconstruct the attitude of instruments and ISP, and we studied how the low-angle lighting impacted this technique. The lessons learned by the direct experience of the LEMS-A3 team at the Exolith lab are being folded into our mission design.

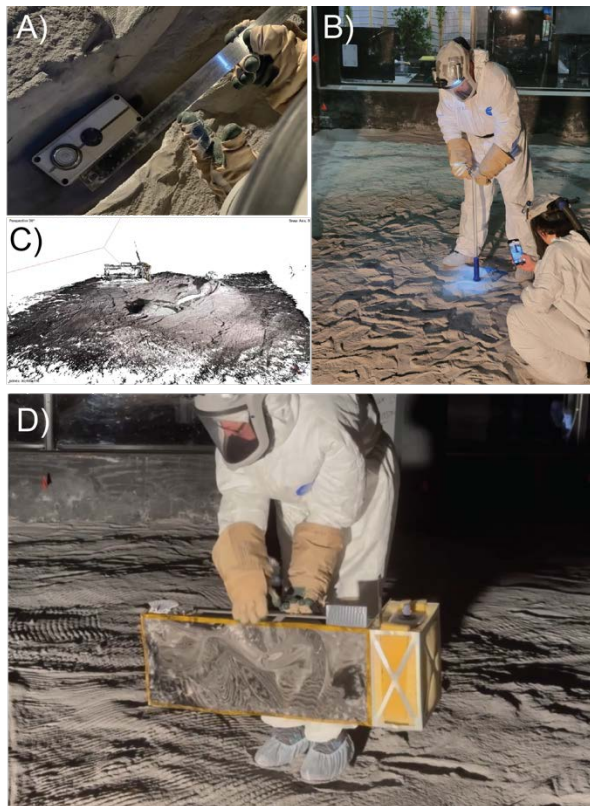


Figure 2. The LEMS-A3 team testing deployment strategies at the UCF Exolith Lab. a) Trenching and burial of the BB instrument. b) Borehole extraction and emplacement of the SP using a AET mockup. c) Photogrammetry reconstruction of the deployment (courtesy of Z. Morse). d) Positioning and leveling of the ISP into the regolith simulant chamber.

Mission Science: All the potential A3 candidate landing sites enable new opportunities for understanding lunar seismology. The South Pole affords us the chance to study seismicity in highlands terrane and ex-

amine farside moonquakes, neither of which were detected in detail by Apollo. The three-month threshold mission and two-year baseline operation of LEMS-A3 will provide a new moonquake catalog that contains impacts, thermal shallow, and deep moonquake activity. The detection and location of 3–4 high signal-to-noise moonquake events, together with the application of single-station data processing techniques developed and advanced during the InSight mission [e.g., 11], will provide new information on the thickness of the lunar crust and properties of the mantle and core. A particular science objective is the structure of the lunar crust and mantle at the margin of the SPA, a basin-forming impact that likely heavily modified and thinned the local crust and deposited proximal ejecta blankets kilometers thick. The formation and modification of the SPA basin seems to have played a fundamental role in the evolution of the lunar surface, so evaluating the crustal thickness and layering, and underlying mantle will provide a basis for understanding the basin formation process and lunar internal evolution [12]. In addition, the LEMS-A3 station will be situated in a region of potentially enhanced volatile abundance in the near surface and perhaps even deeper into the regolith and megaregolith.

Finally, two years of monitoring by LEMS-A3 will provide a long-term assessment of seismic hazards to future sustained human exploration and infrastructure that will be built at the South Polar region. Ground motion could prove a threat to tall towers and structures and/or subsurface dwellings. LEMS-A3 is a technical demonstrator of a low-cost, self-sustaining, long-lived geophysical station at the Moon and could serve as the beginning of a *global* network of geophysical instruments, i.e., in coordination with the Farside Seismic Suite (FSS) [13] and/or a future Lunar Geophysical Network (LGN) [14].

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